Passing Coarse Marginal Checks Can Be Cheap: Persona Mixtures and Imprecise Treatment-Response Estimates in an LLM Persona Panel

Yohei Nakajima
Untapped Capital

Abstract

Large language models are used as synthetic participants, often validated through resemblance to human marginal responses. A fixed panel of sixteen persona-conditioned GPT-4.1 prompts met preregistered broad-reference mean criteria in three of four repeated-game cells. Between-prompt variation was substantial but prior-sensitive. Aggregate continuation-probability contrasts were +0.083 and +0.078, with model-based simultaneous intervals [−0.171, +0.330] and [−0.181, +0.330]. The treatment changed both continuation and its textual representation. Separate representation operations produced large choice shifts without identifying a unique mechanism. External review exposed dependence and multiplicity defects; additive reanalysis preserved the historical record while withdrawing a favored persona-level interpretation. These findings show what coarse validation left imprecisely estimated in one fixed deployment. They establish neither a zero response nor equivalence, human treatment-effect error, or human substitutability. The released archive supports credential-free replay.

1. Introduction

A synthetic participant may produce plausible aggregate choices without supplying precise evidence about the response object an experiment needs. Here, "cheap" denotes evidentiary economy: a validation criterion can be satisfied without requiring that evidence. We examine one fixed sixteen-prompt panel in strategic interaction. Three of four repeated-game condition means passed broad preregistered reference bands, while continuation-probability response estimates remained imprecise.

This question belongs to experimental validation rather than a general verdict on artificial intelligence. Algorithmic fidelity (Argyle et al., 2023), simulated economic agents (Horton, 2023), and behavioral similarity tests (Mei et al., 2024) provide reasons to investigate synthetic participants. Li and Ji (2026) show that statistical realism can diverge from treatment-effect accuracy; Persson et al. (2026) state causal-surrogacy assumptions; Lin et al. (2026) diagnose latent-user drift under intervention. We do not claim to discover that divergence or exhaust its mechanisms.

The closest JESA comparison is Polachek et al. (2025), who compare LLM prisoner's-dilemma choices with a human experiment while varying stakes, order, and framing. Prompt sensitivity and imperfect imitation are already established territory. Our narrower addition is a registered, fixed-panel decomposition of between-prompt composition, within-prompt variation, and represented continuation response, coupled with an explicit dependence and correction record. Unlike their human comparison, our protocol-nonmatched human references cannot identify human treatment-effect error.

Repeated-game prompting and parameter variation also precede this study (Akata et al., 2025; Pal et al., 2026). Strong effect forecasting across human studies is positive evidence for a different use: Ashokkumar et al. (2026) forecast experimental effects, rather than construct the subject-level response surface examined here. Calibration (Hullman et al., 2026) and latent-drift diagnostics can complement our fixed-panel analysis. The later respondent-coverage study of Kumagae and Kaneko (2026), posted after our August preprint,

likewise distinguishes answer marginals from joint respondent profiles; its survey coverage target differs from our strategic-response target. Supplement S1 retains the broader literature and exact versions.

2. Design and estimands

The primary historical deployment used GPT-4.1, temperature 0.7, and outputs capped at sixteen tokens. A minimal subject prompt supplied no reasoning scaffold or game-theory vocabulary. The OpenAI-compatible adapter supplied temperature and `max_tokens=16`; `top_p=1` was specified in the assembly but omitted from the wire. Other sampling penalties inherited provider defaults. No tools or native structured output were used. Environment randomness was seeded; provider generation was not claimed to be seeded.

Before Phase 5 dispatch, a seeded registry generated sixteen complete persona sentences from two age bands and three binary trait dimensions: agreeable/competitive, patient/impulsive, and risk-averse/risk-seeking. Cooperative leaning meant at least two of agreeable, patient, and risk-averse, yielding eight prompts per stratum without reference to behavior. Names and occupations are uncontrolled semantic components. Each complete sentence was prepended to sealed task text. The panel is a fixed construction, not a probability sample of people or a persona generator.

Phase 5 crossed these prompts with four repeated prisoner's-dilemma cells (continuation probabilities .10/.90 $\times$ wording families S2 absent/present), a one-shot label swap, and a Community frame. Primary repeated cells have six complete episodes each. Two seats share an episode; calls, rounds, and seats are not independent subjects. For prompt i, condition d, and episode e, the round-one cooperation outcome is $Y_{ide} \in \{0, 0.5, 1\}$, the fraction of its two seats cooperating. Define $p_i(d) = E[Y_{ide} \mid i, d]$ and the equally weighted fixed-panel mean $\bar{p}(d) = 16^{-1} \sum_i p_i(d)$. The aggregate represented-treatment contrast is $\Delta = \bar{p}(.90) - \bar{p}(.10)$ within each wording family.

The continuation intervention changed the stochastic environment and the sentence communicating it jointly. It identifies response to that represented treatment, not an isolated incentive effect. Reusing explicit prompt text identifies a prompt-indexed contrast; it does not establish a stable latent person's potential outcomes. The deployment, prompt, condition, episode, seat, round, and request levels must remain distinct.

Phase 3 tested the bare instrument; Phase 4 studied representation, labels/payoffs, continuation, adversaries, and monitoring; Phase 5 added the persona panel. Phase 1–2 development is exploratory. Confirmatory predicates preceded their adjudicating data, including sequential X1 registration after Phase 3. Some later design choices were result-informed prospective amendments, and the present sensitivity analyses are post-adjudication. Supplement S2 links the actual seals, dispatch, amendments, adjudications, and reviewer role changes; draft-looking sealed headers are preserved rather than rewritten.

2.1 Uncertainty and validation criteria

We use the episode, retaining uncertainty when observed cells are unanimous. Write $A = \mathbf{1}(Y \geq 0.5)$ and $B = \mathbf{1}(Y = 1)$, so $Y = (A + B)/2$. Simultaneous Clopper–Pearson component intervals project onto condition means. Four component-condition intervals split the error rate by Bonferroni for each contrast. If the projected intervals are $[L_d, U_d]$, the reported contrast interval is $[L_{.90} - U_{.10}, U_{.90} - L_{.10}]$. This conservative discrete small-sample construction uses independent episodes and a pooled component-binomial working model; prompt heterogeneity and the precise coverage domain require care (Supplement S3). It is distinct from treating two seats as independent Bernoulli trials (Clopper and Pearson, 1934).

Three composition views are intentionally separated. Finite-opportunity-corrected plug-in estimates and conditional episode bootstraps describe the recorded panel; unanimous cells remain degenerate in that bootstrap. Symmetric Dirichlet priors on each prompt-cell's three outcome probabilities propagate sparse-cell uncertainty but are prior-sensitive. A two-stage prompt-plus-episode bootstrap changes the target toward a hypothetical persona generator. These are different conditioning assumptions, not independent replications of one estimate.

Table 1 distinguishes the validation criterion from the desired inference. Broad condition-specific bands $[\ell_d, u_d]$ imply only $\Delta \in [\ell_1 - u_0, u_1 - \ell_0]$. Exact matching of both means would fix their difference; our point concerns slack in coarse criteria. The law of total variance and classical coupling results explain why marginal moments do not determine between/within composition or cross-condition coupling (Supplement S3); these applications are not new probability theorems.

Target	Criterion or estimate	What it supports
Marginal level	Reference band [.36,.63]; 3/4 means inside	Passage of a broad historical check
Composition	Plug-in 85–96%; Dirichlet medians depend on prior	Substantial prompt-indexed variation in this panel
Aggregate response	+.083 / +.078; simultaneous intervals about −.18 to +.33	Imprecise represented-treatment response
Latent-person response	No invariance test	Unidentified
Human causal accuracy	Protocol-nonmatched references	Unidentified

Table 1. Validation targets, evidence, and inference limits. Composition percentages are between-prompt shares of episode-level variation. Neither the bands nor the variance heuristic is an equivalence margin.

3. Results

3.1 Marginal fit and composition

Repeated-cell cooperation means were 0.349, 0.427, 0.432, and 0.505; the sole reference-band miss was 0.011 below the lower bound. Cooperative-leaning minus defect-leaning prompt-bundle differences were 0.510–0.719. They are not isolated causal trait effects.

Between-prompt plug-in shares were 85.5%, 96.1%, 88.8%, and 90.2%. Fixed-panel Jeffreys Dirichlet$(0.5, 0.5, 0.5)$ posterior medians were 63.1%, 70.5%, 66.1%, and 66.6%; respective 95% intervals were [49.4%,74.5%], [57.3%,81.3%], [52.7%,77.1%], and [52.7%,77.7%]. A symmetric-prior sweep yielded medians 74.8–82.5% at $\alpha = .25$ and 47.1–53.5% at $\alpha = 1$. Thus substantial composition persists, but a share exceeding one-half is prior-dependent (Figure 1).

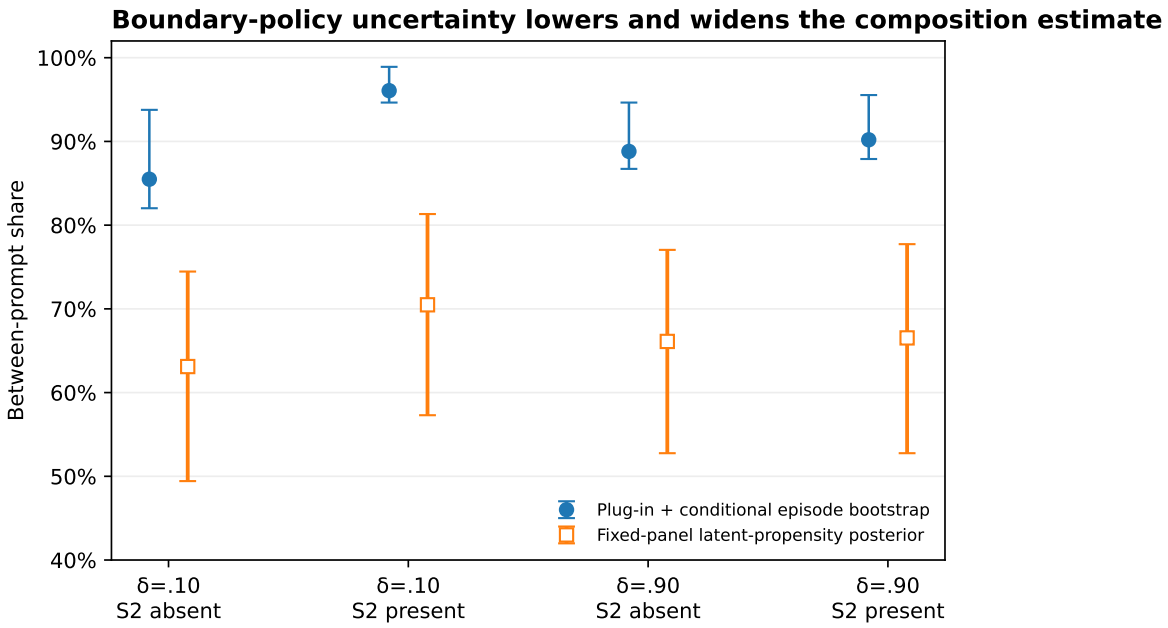


Figure 1. Plug-in/conditional-bootstrap and Jeffreys fixed-panel composition estimates. The former conditions on recorded corner concentration; the latter shrinks sparse cells inward. Uniform-prior medians reported in text straddle one-half. These estimates are not interchangeable.

P5-1b compared corrected dispersion with heuristic thresholds derived from Dal Bó and Fréchette's (2011) human strategy frequencies. Those experiments differ in continuation, payoffs, incentives, assignment, and experience. The frozen ratio .75 is a tolerance, not a human-equivalence margin. The historical dispersion checkpoint is permissive; neither its passage nor matched-looking SD establishes human variance equivalence.

The binary interiority census is also method-sensitive. In the registered restricted set, the frozen seat rule classified 3/32 cells interior, the episode-exact rule 2/32, and Jeffreys sensitivity 5/32. Across all six conditions, the corresponding counts are 14/96, 11/96, and 19/96. Six episodes and discrete outcomes make threshold labels unstable; continuous estimates are more informative.

3.2 Response and representation

Continuation contrasts were +0.083 (S2 absent) and +0.078 (S2 present). Under the pooled component-binomial working model, simultaneous 95% intervals were [−0.171,+0.330] and [−0.181,+0.330]. These are exact descriptive differences for the archived panel but imprecise estimates of its repeated-sampling propensities. They establish neither equivalence, zero response, nor a narrow upper bound (Figure 2). A coarse validation design did not require this response object to be precisely estimated.

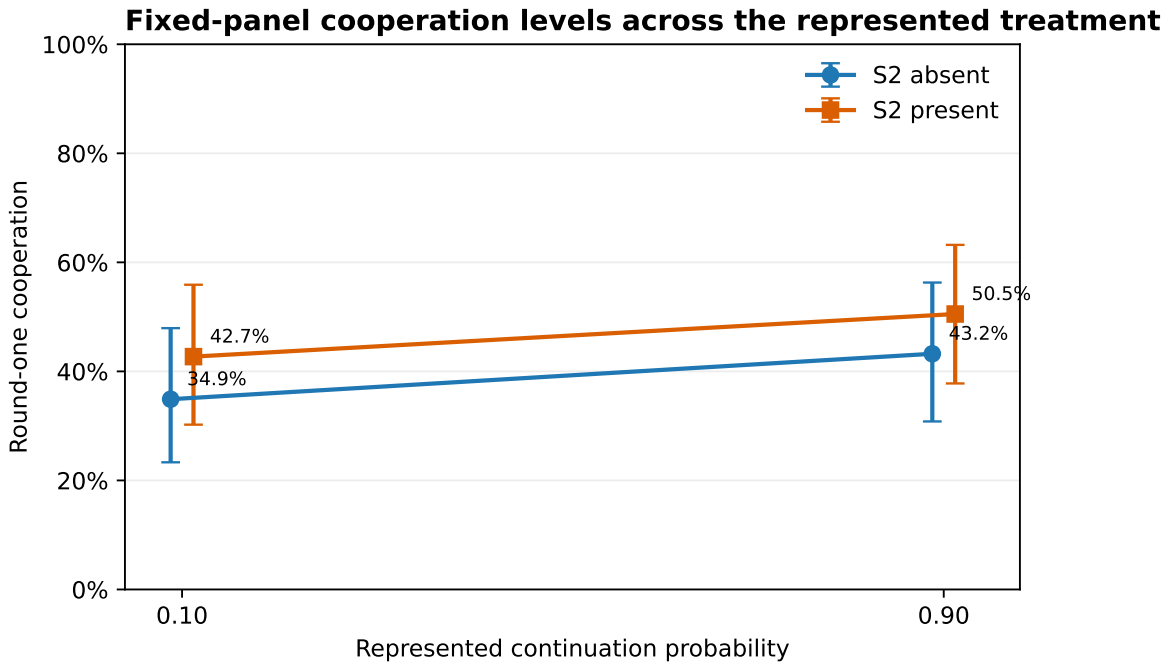


Figure 2. Cooperation levels and episode-based condition intervals under the pooled component-binomial working model. Lines orient the eye; contrast intervals in text use simultaneous component projection. Process and communicating text change together.

For the bare configuration, a screened and held-out S2 wording-and-position replacement shifted cooperation from 0/40 to 37/40 seat choices. Wording and position were one operation. By contrast, ordinary one-shot wording had a registered main effect of +0.0063 (SE .0210; Holm-adjusted $p = 1.00$). Announcing and implementing repeated interaction produced larger changes, including ceiling-confounded cells; those ceilings cannot identify a continuation slope.

With canonical payoffs held fixed, swapping "Cooperate" and "Defect" made the bare configuration always choose the dominated role labeled "Defect." This demonstrates label-linked control in that cell, but learned game associations and intrinsic lexical valence are not separated. Adding a persona reversed the bare choice; no format-matched neutral prefix was tested, and label and payoff then favored the same option. Neither operation isolates a universal semantic mechanism.

P5-2 pooled 90/704 seat decisions across 352 episodes, equivalently an episode mean of 45/352=.128. The principal dependence-aware sensitivity, a stratified bootstrap over forty prompt-by-conflict clusters, gave [.071,.189]. Fixed-panel Dirichlet aggregation moved from median .138 at $\alpha = .10$ to .205 at $\alpha = 1$, whose interval [.182,.231] crosses the registered .20 boundary. The historical verdict remains visible; Bayesian proximity to its boundary is not prior-robust corroboration. Repeated conflict subcells were mixed; the pooled dominance classification is carried by the word/payoff-confounded swap cell.

3.3 Multiplicity and the withdrawn persona interpretation

The frozen P5-3(a) search selected persona p13 after searching multiple persona-by-wording candidates without prospective family control. Its apparent capability finding is withdrawn. Three post-adjudication 200,000-permutation analyses dynamically reapply their eligibility gates in every draw. Add-one family-wise p values and Monte Carlo 95% intervals are .059230 [.058194,.060268] under the historical seat gate;

.043455 [.042561,.044353] under the percentile episode-bootstrap gate; and .773206 [.771363,.775039] for the largest eligible candidate p05 under the conservative exact gate, where p13 is ineligible.

The favorable bootstrap variant cannot create prospective confirmation; none was sealed before computation. Conversely, the exact gate cannot decisively disconfirm the capability. At six episodes, its eligible compositions permit a maximum slope .333; the archived family's estimated minimum attainable family-wise tail probability is .075040 [.073884,.076198], above .05. P13 is a prospective replication target, not an established finding or a decisive negative. No new study is needed for the present bounded conclusion.

No study-wide alpha allocation was registered across P5 predicates and X1. Registered block-specific controls, including Holm-controlled adversary tests, retain their own scope. The complete supplementary record retains favorable and unfavorable variants, dead predictions, invalidations, and secondary findings. Mechanical predicate enforcement cannot ensure construct validity or an appropriate family.

4. Implications and limitations

Validation should name the target, intervention representation, dependence unit, and uncertainty required for the intended use. A panel can pass a coarse level check while leaving treatment response loosely bounded. This is a limitation of the criterion demonstrated in one fixed deployment, not evidence that matched human responses would be different.

The model version and sixteen bundled prompts bound generalization. Prompt pairing does not establish latent-person invariance; no new human comparison, wider persona sample, factorial mechanism control, or endpoint refresh is claimed. Gemini observations are descriptive because its unversioned endpoint drifted. Temperature interactions outside the frozen plan are exploratory. More episodes may improve within-prompt precision, but a wider persona-generator claim requires a declared sampling design.

Auditability enabled correction rather than self-validation. A sealed pre-data discussion originally interpreted p13 as an incentive-response existence result; review exposed dependence, family-error, and construct defects. The original discussion, mechanical verdicts, seals, and amendment history remain unchanged. One AI reviewer later helped specify post-adjudication analyses; this role transition and later independent checks are disclosed in Supplement S2. The human author retains responsibility. These records preserve the distinction between registered predictions, result-informed prospective changes, descriptive observations, and later reinterpretation.

5. Data, code, and declarations

The immutable evidence baseline is repository commit 62ab90beff1d291b069c87f13b6b6b3495171ddd; the distinct public preprint is arXiv:2608.00979v1. Supplement S4 and the separate Data and Instructions appendix link the code, registrations, complete prompts, data dictionary, licenses, and audit receipts. The archive contains 4,916 confirmatory runs: 320 Phase 3/X1 LLM, twenty deterministic baselines, 2,864 Phase 4, and 1,712 Phase 5. Three legacy diagnostics raise the verifier boundary to 4,919 runs, of which 4,899 are LLM runs; runs are not provider calls.

Phase 3 lacks the richer response IDs and request-body hashes of Phases 4–5. The latter contain 30,421 request events and 30,397 response events; twenty-four failed-run partials are excluded from completed analyses but retained in accounting. Snapshot checksums and timestamps detect changes relative to publication; they are not provider attestation or proof of immutability from receipt. Replay reconstructs archived behavior without live models and cannot establish human validity or adequate power.

Author affiliation/contact/ORCID, contributions, actual funding/support, interests, human-data determination, rights, full AI assistance history, and current journal history await author confirmation in a separate worksheet. This review candidate must not be submitted with unresolved declarations. Subject-model apparatus and AI assistance are distinct: archived subject models generated observations; AI coding, writing, analysis, and adversarial-review roles require the author-approved disclosure. AI systems are not authors.

References

Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., and Schulz, E. (2025). Playing repeated games with large language models. *Nature Human Behaviour, 9*, 1380–1390. https://doi.org/10.1038/s41562-025-02172-y

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis, 31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Ashokkumar, A., Hewitt, L., Ghezae, I., and Willer, R. (2026). Large language models can predict the results of social science experiments. *Nature, 656*, 115–122. https://doi.org/10.1038/s41586-026-10742-x

Clopper, C. J., and Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika, 26*(4), 404–413. https://doi.org/10.1093/biomet/26.4.404

Dal Bó, P., and Fréchette, G. R. (2011). The evolution of cooperation in infinitely repeated games: Experimental evidence. *American Economic Review, 101*(1), 411–429. https://doi.org/10.1257/aer.101.1.411

Horton, J. J. (2023). Large language models as simulated economic agents: What can we learn from Homo Silicus? NBER Working Paper 31122, original April 2023 version. https://www.nber.org/system/files/working_papers/w31122/revisions/w31122.rev0.pdf. https://doi.org/10.3386/w31122

Hullman, J., Broska, D., Sun, H., and Shaw, A. (2026). This human study did not involve human subjects: Validating LLM simulations as behavioral evidence. arXiv:2602.15785v1. https://arxiv.org/abs/2602.15785v1. https://doi.org/10.48550/arXiv.2602.15785

Kumagae, Y., and Kaneko, M. (2026). Silicon samples match the answers but not the respondents. arXiv:2610.30900v1. https://arxiv.org/abs/2610.30900v1. https://doi.org/10.48550/arXiv.2610.30900

Li, Z., and Ji, F. (2026). Statistical realism is not evidence that LLMs can estimate treatment effects in social science experiments. arXiv:2604.02458v3. https://arxiv.org/abs/2604.02458v3. https://doi.org/10.48550/arXiv.2604.02458

Lin, V., Yun, T., Matarić, M. J., Canny, J., Gretton, A., and D'Amour, A. (2026). The illusion of intervention: Your LLM-simulated experiment is an observational study. arXiv:2605.20767v1. https://arxiv.org/abs/2605.20767v1. https://doi.org/10.48550/arXiv.2605.20767

Mei, Q., Xie, Y., Yuan, W., and Jackson, M. O. (2024). A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences, 121*(9), e2313925121. https://doi.org/10.1073/pnas.2313925121

Pal, S., Mallela, A., Hilbe, C., Pracher, L., Wei, C., Fu, F., Schnell, S., and Nowak, M. A. (2026). Strategies of cooperation and defection in five large language models. arXiv:2601.09849v1. https://arxiv.org/abs/2601.09849v1. https://doi.org/10.48550/arXiv.2601.09849

Persson, J., Schultzberg, M., and Ankargren, S. (2026). Statistical foundations of LLM-based A/B testing: A surrogacy framework for human causal inference. arXiv:2606.17165v3. https://arxiv.org/abs/2606.17165v3. https://doi.org/10.48550/arXiv.2606.17165

Polachek, S. W., Romano, K., and Tonguc, O. (2025). Homo-silicus: not (yet) a good imitator of homo sapiens or homo economicus. *Journal of the Economic Science Association, 11*(2), 261–269. https://doi.org/10.1017/esa.2025.10023